

(b) (i) $x = 138$
 $y = 40$

(ii) Energy released

$$= (235.0439 + 1.0087 - 137.9603 - 97.9197)(1.66 \times 10^{-27})(3.00 \times 10^8)^2$$

$$= 2.58 \times 10^{-11} \text{ J}$$

(iii) Electromagnetic radiation (eg. γ -radiation)

(iv) 1. From conservation of linear momentum,

$$0 = m_{Te}v_{Te} + m_{Zr}v_{Zr}$$

$$\left| \frac{v_{Zr}}{v_{Te}} \right| = \frac{m_{Te}}{m_{Zr}} = \frac{138}{98}$$

$$2. \quad \frac{KE_{Zr}}{KE_{Te}} = \frac{\frac{1}{2}m_{Zr}v_{Zr}^2}{\frac{1}{2}m_{Te}v_{Te}^2} = \frac{m_{Zr}}{m_{Te}} \left(\frac{v_{Zr}}{v_{Te}} \right)^2 = \left(\frac{98}{138} \right) \left(\frac{138}{98} \right)^2 = \frac{138}{98}$$

$$(v) \quad \frac{1}{2}m_{Zr}v_{Zr}^2 = \frac{138}{138 + 98} \times 2.3 \times 10^{-11}$$

$$\frac{1}{2}(97.9197 \times 1.66 \times 10^{-27})v_{Zr}^2 = 1.345 \times 10^{-11}$$

$$v_{Zr} = 1.29 \times 10^7 \text{ m s}^{-1}$$

Note: Candidates should be advised to consider whether their answers are reasonable.

(vi) 1. The total momentum of the system before the nuclear reaction was zero, i.e. the momentum of the slow neutron was negligible.

2. The momentum of the γ -ray photons emitted after the reaction was negligible.